Pterodactyl Defense

ECO A40-B27 | 33 named lines | Flank openings (English, Reti, Bird, KIA)

In one sentence: Flexible flank openings control the center with pieces and avoid early theoretical battles.

Memory pattern: Fianchetto, control d5 and e5, then pick the center.



Main line: 1.Nf3 c5 2.c4 g6 3.d4 Bg7 4.e4 Qa5 5.Nc3 d6

Key ideas

1. Control d5 and e5 with pawns on c4 and pieces on f3 and c3.
2. Fianchetto the king's bishop (g3, Bg2) for long-diagonal pressure.
3. Choose a center after seeing Black's setup: d4 or e4.
4. Transpose to Queen's Gambit, Catalan or Sicilian-style structures.
5. English: reversed Sicilian ideas with extra tempo.

Reading the diagram

Blue arrows = thematic plan moves for the side to move. Green arrow = engine's best move.
Engine at the end of the main line: **+0.61** (slightly better for White, depth 17). Best: d5.

The five coach questions

1. Ideal pawn structure?	Control d5/e5 with pieces and keep the centre flexible (Reversed Sicilian, Reti, King's Indian Attack). Transpositions are common.
2. Where does my king go?	Castle short; g3/Bg2 is a safe shelter.
3. Best minor pieces?	Bg2 and Bb2 on the long diagonals, knights c3/f3, breaks with d4/e4.
4. Main middlegame plan?	Choose a centre after seeing Black's setup; expand with b4 or d4; attack with e4/Ne5.
5. What does my opponent want?	White wants a favourable centre; Black wants to claim the centre with ...e5/...d5 or symmetrical counterplay.

The main line, move by move

Most common continuation in the database: 1.Nf3 c5 2.c4 g6 3.d4 Bg7 4.e4 Qa5 5.Nc3 d6

Move	Why it is played
1. Nf3	Develops the knight to f3.
1... c5	Symmetry and central pressure.
2. c4	Control d5. Pawn move that supports the center or prepares a break.
2... g6	Fianchetto. Gains space on the wing.
3. d4	Central break. Occupies the center with a pawn.
3... Bg7	Long diagonal. Fianchetto: the bishop takes the long diagonal.
4. e4	Take the center. Occupies the center with a pawn.
4... Qa5	Queen move: puts pressure on a target (watch for tempo-gaining attacks).
5. Nc3	Develops the knight to c3.
5... d6	Pawn move that supports the center or prepares a break.



After 1.Nf3 c5 2.c4 g6



After 4 g6 3.d4 Bg7 4.e4 Qa5

Key variations

The most important named branches (ranked by how much theory lies beneath them). Diagram arrows: blue = plan moves, green = Stockfish.



Position after the line

Queen Pterodactyl A40 | 1 lines

`1.d4 g6 2.c4 Bg7 3.Nc3 c5 4.e3`

Stockfish: **+0.21** (equal, depth 16). Best next move: cxd4.
What to look for: e5 (take the center); Nf6 (develop); d5 (hit c4)
Sub-variations: Quiet Line

Expert and coach view

What a coach would tell you

Flank openings are about flexibility, not speed. Make sure you know the main transpositions and what structures you want, then choose moves accordingly.

Why play it

Flexible flank openings control the center with pieces and avoid early theoretical battles.

Typical mistakes

- Being too passive and letting Black take the center for free.
- Playing d3 and e4 without a plan for the queenside pawns.
- Neglecting development while chasing a fianchetto.

Typical tactics and traps

- Early Qb3 or Qa4 pins that win the c4 or d5 pawn if Black is careless.
- Tactics on the long diagonal against b7/g7.

Endgame and middlegame outlook

Balanced, flexible endgames with minor advantages for the better-placed bishops.

Study checklist

[ ] Play the main line on a board and say each move's purpose aloud.

[ ] Learn the five coach questions by heart; answer them without notes.

[ ] Drill two sub-variations until the moves feel automatic.

[ ] Play three games with the opening, then review where you left theory.

[ ] Re-read the traps page the day before a tournament.

The opening tree

Every named line in this family as a tree. Gray moves are inherited from the parent; green after the arrow are the new moves that define the variation. Large green dots mark branches with 10 or more lines.

- **Pterodactyl Defense (all lines)** A42
  `1.Nf3 c5 2.c4 g6 3.d4 Bg7 4.e4 Qa5 5.Nc3 d6`
  - **Central** A40
    - **Benoni Pterodactyl** A40
      `1.d4 g6 2.c4 Bg7 3.e4 c5 4.d5 Qa5`
    - **Anhanguera** B06
      `1.e4 g6 2.d4 Bg7 3.c4 c5 4.Nc3 d6 5.Be3 Qa5`
    - **Benoni Quetzalcoatlus** B06
      `1.e4 g6 2.d4 Bg7 3.c4 c5 4.d5 d6 5.Nc3 Qa5`
    - **Bogolubovia** A42
      `1.d4 g6 2.c4 Bg7 3.e4 d6 4.Nc3 c5 5.Nge2 Qa5`
    - **Quetzalcoatlus Gambit** B06
      `1.e4 g6 2.d4 Bg7 3.c4 c5 4.Nc3 d6 5.dxc5 Qa5`
    - **Benoni Beefeater Pterodactyl** A40
      `1.d4 g6 2.c4 Bg7 3.Nc3 c5 4.d5 d6 5.e4 Bxc3 6.bxc3 Qa5`
  - **Eastern** B06
    - **Anhanguera** B06
      `1.e4 g6 2.d4 Bg7 3.Nc3 c5 4.Be3`
    - **Benoni** B06
      `1.d4 g6 2.e4 Bg7 3.Nc3 c5 4.d5`
    - **Benoni Pterodactyl** B06
      `1.d4 g6 2.Nc3 Bg7 3.e4 c5 4.d5 Qa5`
    - **Rhamphorhynchus** B06
      `1.e4 g6 2.d4 Bg7 3.Nc3 c5 4.dxc5 Qa5`
    - **Benoni Pteranodon** B06
      `1.e4 g6 2.d4 Bg7 3.Nc3 c5 4.d5 Bxc3 5.bxc3 Qa5`
    - **Pteranodon** B06
      `1.e4 g6 2.d4 Bg7 3.Nc3 c5 4.dxc5 Bxc3 5.bxc3 Qa5`
  - **Fianchetto** A40
    - **King Pterodactyl** B06
      `1.e4 g6 2.d4 Bg7 3.g3 c5 4.Nf3 Qa5`
    - **Queen Benoni Pterodactyl** A40
      `1.d4 g6 2.c4 Bg7 3.Nc3 c5 4.d5 Qa5`
    - **Queen Pterodactyl** A40
      `1.d4 g6 2.Nf3 Bg7 3.g3 c5 4.Bg2 Qa5`
    - **Rhamphorhynchus** B06
      `1.e4 g6 2.d4 Bg7 3.g3 c5 4.dxc5 Qa5`
    - **Queen Pteranodon** A40
      `1.d4 g6 2.c4 Bg7 3.Nc3 c5 4.d5 Bxc3 5.bxc3 Qa5`
  - **Sicilian** B06
    `1.e4`
    - **Anhanguera** B27
      `1.e4 -> 1...c5 2.Nf3 g6 3.d4 Bg7 4.Nc3 Qa5 5.Be3`
    - **Benoni Gambit** B27
      `1.e4 -> 1...c5 2.Nf3 g6 3.d4 Bg7 4.Nc3 Qa5 5.d5`
    - **Rhamphorhynchus** B27
      `1.e4 -> 1...c5 2.Nf3 g6 3.d4 Bg7 4.dxc5 Qa5 5.Nc3`
    - **Siroccopteryx** B06
      `1.e4 -> 1...g6 2.d4 Bg7 3.Nc3 c5 4.Nf3 Qa5 5.Bc4`
    - **Unpin** B27
      `1.e4 -> 1...c5 2.Nf3 g6 3.d4 Bg7 4.Nc3 Qa5 5.Bd2`
    - **Quetzalcoatlus** B06
      `1.e4 -> 1...g6 2.d4 Bg7 3.Nc3 c5 4.Nf3 Qa5 5.Be2 d6`
    - **Pteranodon** B27
      `1.e4 -> 1...c5 2.Nf3 g6 3.d4 Bg7 4.dxc5 Qa5 5.Nc3 Bxc3 6.bxc3`

- **Western** B06
  1.e4
  - **Anhanguera** B06
    1.e4 -> 1...g6 2.d4 Bg7 3.Nf3 c5 4.Be3 Qa5
  - **Pterodactyl** B27
    1.e4 -> 1...c5 2.Nf3 g6 3.c3 Bg7 4.d4 Qa5
  - **Rhamphorhynchus** B27
    1.e4 -> 1...c5 2.Nf3 g6 3.d4 Bg7 4.dxc5 Qa5
  - **Siroccopteryx** B06
    1.e4 -> 1...g6 2.Nf3 Bg7 3.d4 c5 4.Bc4 cxd4 5.Nxd4 Qa5
- **Austrian** B06
  1.e4 g6
  - **Austriadactylus Western** B06
    1.e4 g6 -> 2.d4 Bg7 3.f4 c5 4.Nf3 Qa5
  - **Grand Prix Pterodactyl** B06
    1.e4 g6 -> 2.Nc3 Bg7 3.f4 c5 4.Nf3 Qa5
  - **Pteranodon** B06
    1.e4 g6 -> 2.d4 Bg7 3.f4 c5 4.c3 Qa5
- **Queen Pterodactyl** A40
  1.d4 g6 2.c4 Bg7 3.Nc3 c5 4.e3
  - **Quiet Line** A40
    1.d4 g6 2.c4 Bg7 3.Nc3 c5 4.e3

One-page memory sheet

Move-order pattern

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1.Nf3 c5 2.c4 g6 3.d4 Bg7 4.e4 Qa5 5.Nc3 d6
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Plan chain

White	c4 -> Nf3 -> g3 -> Bg2 -> O-O
Black	e5 -> Nf6 -> d5 -> c5 -> g6

Three rules to remember

1. Fianchetto, control d5 and e5, then pick the center.

2. Choose a centre after seeing Black's setup; expand with b4 or d4; attack with e4/Ne5.

3. White wants a favourable centre; Black wants to claim the centre with ...e5/...d5 or symmetrical counterplay.

Pawn structure in one line

Control d5/e5 with pieces and keep the centre flexible (Reversed Sicilian, Reti, King's Indian Attack). Transpositions are common.